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Developing Capacity for Leading Change in Rural Educational Contexts through Simulation

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Developing Capacity for Leading Change in Rural Educational Contexts through Simulation*

Abstract

Building the leadership capacity of educators is a critical component of educational change. Rural school divisions experience unique contextual challenges related to the development of leadership capacity. Promising practices, such as the use of game board simulations like "Leading Professional Learning: Building Capacity for Sustained Effective Practice" (Stiles, Mundry & Bershad, 2016), have the potential to develop leadership potential, and spark educational innovation. Informed by critical constructivism, this qualitative multiple case study sought to answer the following question: How does participation in a professional development board game simulation serve as a catalyst for developing leadership capacity in rural educators? Data sources included online questionnaires, semi-structured interviews with participants, PD artifacts, and recorded debrief sessions between the PD facilitators/researchers. Findings from the study suggested that the game-like structure of the simulation, which required teams to work together to achieve positive outcomes and outpace their rivals, enhanced motivation and engagement, while leading to new insights about leadership and the change process. Such insights were developed within, and in relation to the rural contexts of the educators involved, connecting individual and contextual knowledge with broader theories about educational change.

This research study emerged organically from the experiences of both researchers. Smith's (2014) doctoral research focused on developing the leadership capacity of teacher leaders. As a consequence, she was invited to facilitate a professional learning series on teacher leadership and help instruct a year long field-led graduate course titled Change Leadership, for the Manitoba Teachers' Society (MTS). One of the MTS Staff Officers, Fullan Kolton, had encountered the PD Simulation at Lucy West's Summer Institute, and suggested it be included in the two day introduction to the Field-Led Course on Change Leadership.

"Leading Professional Learning: Building Capacity for Sustained Effective Practice" (Stiles, Mundry & Bershad, 2016) is a WestEd (WestEd.com) learning tool designed to simulate two years in the life of a school-based Professional Development Committee trying to move a specific group of individual staff members on a game board path towards developing competency in science instruction. Teams select specific strategies that help teachers accumulate expertise and move towards full implementation of the Science initiative.

The initial experience of using the WestEd PD simulation as part of the Change Leadership Course was very positive, and spurred subsequent use in other contexts. Following this experience Smith facilitated professional learning networks through the Manitoba Rural Learning Consortium (mRLC), and in each network, the PD simulation was an integral component. Participants continued referring to their simulation experience throughout the year-long networks, suggesting that it had a strong impact on their thinking about change leadership.

Skyhar's (2018) doctoral research focused on "the problem of *how* rural school divisions and teachers might go about creating models of PD that lead to the provision of effective and meaning PD for teachers, and that mitigate the challenges faced in providing effective PD locally" (p. 9). The study, which investigated a teacher professional development model

implemented in a very small rural school division in southern Manitoba, illuminated several challenges rural divisions and teachers face with regards to teacher PD. Among these challenges, was the limited capacity of very small rural divisions due to the small number of educators working within them, and the fact that they often have fewer formal leadership positions and opportunities to develop leadership capacity.

Commensurate with the above experiences, both researchers joined an education faculty at a university in western Manitoba. Common interests in professional learning and leadership development in rural contexts, led to conversations about how to strengthen the leadership capacity of rural teachers. Two sets of the PD simulation were purchased for the university and during 2017-2018 the researchers co-facilitated the simulation with different groups including undergraduate and graduate students, and faculty members. Impressed with the high level of engagement and interest from the range of pre-service and in-service educators, the researchers decided to inquire more deeply into its impact through this research study.

Purpose of the Study

The study that was designed by the researchers focused on the following question: How does participation in a professional development board game simulation serve as a catalyst for developing leadership capacity in rural educators? The development of leadership capacity, the rural context, and the PD simulation game itself were all identified as critical elements of the study.

Purpose in relation to leadership. School leadership, once considered the responsibility of vice-principals and principals, is now more frequently shared amongst staff members. Informal and formal teacher leaders (Lambert, Zimmerman & Gardner, 2016; Katzenmeyer & Moller, 2009; Killion et al., 2016; Levin & Schrum, 2017; York-Barr & Duke, 2004) participate

in distributed leadership (Leithwood & Mascal, 2008) within complex school environments. Teacher leadership develops not only through acquiring new knowledge, skills and dispositions, but also through engaging in leadership activities (Carver, 2016; Cortez-Ford, 2008; Judkins, 2014; Sinha & Hanuscin, 2017). When teachers are treated as leaders by others and are given leadership responsibilities, they begin to identify as teacher leaders, and are more likely to embrace the role of change agent (Calvert, 2016).

Lambert (2003) states that all teachers have the right and the responsibility to provide leadership in school communities. Her vision of "broad-based skillful participation in acts of leadership" (Lambert, 1998), requires schools and teachers to develop leadership capacity. High leadership capacity organizations demonstrate a shared vision, data-informed decisions, collaborative roles, collective responsibility, reflective practice and improved student learning (Lambert, 1998; 2003; Lambert, Zimmerman & Gardner, 2016).

To develop leadership capacity, teachers must have access to quality professional learning opportunities that focus on acquiring the knowledge, skills and dispositions of a change agent (Calvert, 2016; Campbell et al., 2016; Muijs & Harris, 2003). To facilitate change in school systems, educators must understand a school's culture, the change process, how to simulate growth, and how to respond to resistance and conflict (Levin & Schrum, 2017). Learning opportunities that expose teachers to this content and provide realistic opportunities for application of that knowledge, have the potential to strengthen leadership capacity and foster educational change.

Purpose in relation to rural education. Rural educators work within unique contexts that require careful consideration when it comes to professional development, system change, and educational reform. Rural teachers and administrators often have many hats to wear, taking

on multiple roles within their local contexts (Forner, Bierlein-Palmer, & Reeves, 2012; Glover et al., 2016; Hardré, 2009; Newton & Wallin, 2013; Starr & White, 2008; Wallin, 2008). In addition, many rural educators are required to work outside areas for which they were trained, creating a situation in which there is a significant need for professional development and capacity-building (Harmon, Gordanier, Henry, & George, 2007; MASS & MAST, 2006; Tytler, Symington, Darby, Malcolm, & Kirkwood, 2011). The heavy workloads that result from such situations are often compounded by limited access to PD opportunities (particularly those relevant to their contexts) and significant professional isolation due to geographical distances, making it difficult for rural educators to access the supports they need (Hardré, 2009; Harmon et. al, 2007; Howley & Howley, 2005; MASS & MAST, 2006; Tytler et. al, 2011; Wallin, 2008).

Rural school divisions also face significant challenges in relation to the provision of professional development and their capacity to engage in systemic change. Small budgets, declining enrolments, high fixed costs in areas such as salaries and bussing, and inequities in rural funding models pose financial challenges for divisions that necessitate difficult choices, particularly when making decisions about discretionary funding areas such as professional development expenditures (Harmon et. al, 2007; Howley & Howley, 2005; NADC, 2010; Starr & White, 2008; Wallin, 2008, 2009). Increased costs for travel and accommodation, due to geographic distance, also pose challenges for rural divisions, both in relation to internal initiatives, and in terms of providing access to external professional development for educators (Glover et. al, 2016; Harmon et. al, 2007; MASS & MAST, 2006; Wallin, 2009). Finally, the lack of access rural divisions have to pools of qualified substitute teachers, and the small number of formal leadership positions that exist within rural contexts profoundly impact the ability of rural divisions to provide access to effective professional development, further limiting their

capacity to engage in system change and educational reform (Glover et. al, 2016; Harmon et. al, 2007; Skyhar, 2018).

In order for those working in rural contexts to engage in growth, change, and educational reform, attention must be paid to the unique contextual needs of rural educators and organizations, including the challenges they face. Attention must also be paid to the complexity involved in initiating change in rural places, including ways of nurturing and developing local capacity for leading such change.

The potential of PD simulation. PD simulation tools (such as the one created by WestEd) have the potential to act as mediating tools (McCullagh, 2012) for participants to socially construct understandings about teaching, learning, and leadership. Within a social context, participants are prompted to not only think about how to be leaders within the simulated context, but also to make connections between the new understandings they construct through their collaborative participation and their own experiences in rural places. The research conducted examined the experiences of several groups of rural educators that engaged in the WestEd board game simulation in order to determine how their participation served as a catalyst for developing leadership capacity both during the simulation, and beyond in their rural contexts.

Theoretical Framework

This research study was informed by the framework of critical constructivism (Kincheloe, 2005), which has three component parts: critical theory, social constructivism and complexity theory. Critical theory (Apple, 2004; Giroux, 2004; McLaren, 2009) relates to issues of who has power in society and how such privileges are perpetuated. Due to geographic isolation, rural educators have historically had limited access to professional learning opportunities to develop their leadership understanding and skills; this leadership development

opportunity delivered in a rural centre for rural educators aimed to overcome this obstacle. Social constructivism (Dewey, 1938; Vygotsky, 1979) asserts that people are social beings who create understanding and build meaning through discourse with others. Throughout the simulation participants engage in collaborative dialogue, analysis and decision making with their team members allowing them to gain deeper insights into the learning experience than if they participated alone. Complexity theory (Kincheloe & Berry, 2004) focuses on the self-organizing and emergent nature of diverse ecosystems. Schools consist of unique combinations of individual educators and systemic change requires the intense interaction and interdependence of these multiple perspectives, plus effective planning processes and levels of decision-making unique to each context; "In complexity theory ... no part controls the behaviour of the system as a whole … the relationships between the parts produces systemic qualities that are difficult to anticipate" (Kincheloe & Berry, 2004, pp. 36-37). Through playing the simulation board game participants are exposed to the complexities of individual, school and system change and the intricacies of fostering innovation in ever-changing diverse and human contexts.

Methods

As previously mentioned, the guiding question for the research study was the following: How does participation in a professional development board game simulation serve as a catalyst for developing leadership capacity in rural educators? In order to answer the question, a multiple case study was designed to look at the experiences of a variety of groups of rural educators who participated in PD workshops that used a board game simulation titled "Leading Professional Learning: Building Capacity for Sustained Effective Practice" (Stiles et al., 2016) as a mediating tool for learning. Using existing contacts and word of mouth, the researchers offered to run PD sessions for educators in a variety of rural contexts in Manitoba (school divisions, schools,

teacher associations, etc.) The PD workshops required educators to (1) play the board game in teams of 3-5 players, (2) deconstruct their experiences playing the game/simulation, and (3) identify connections between their participation in the game/simulation and their own educational contexts. Workshop participants were also asked to think about what they had learned from the experience, and how it would potentially inform their future actions.

Participants were recruited for the study at PD workshops facilitated by the researchers. All of the educators participating in the PD workshops in each context were invited to be participants in the research study via letters of invitation. Those choosing to be participants in the study indicated their consent by accessing and filling in an online questionnaire after the workshop. The online questionnaire also provided participants with an opportunity to participate in a follow-up, recorded, semi-structured interview with the researchers approximately two months after their workshop experience to reflect on the impact of their workshop experience on their learning and leadership capacity.

While the study is ongoing, this paper reports on the findings related to the initial group of educators who engaged in the PD workshop (the first case). Stake (2006) describes the relationship between the single case and a broader multiple case study in the following way: "A researcher may spend a long or short time on a case, but works vigorously to understand each particular case (one case at a time)" (p. 1). This paper does just that, examining the experiences of the educators in the first case study in detail, with the eventual goal of looking at patterns across cases.

Yin (2009) describes the scope of case study as a methodology in the following way: "A case study is an empirical inquiry that investigates a contemporary phenomenon in depth and within its real-life context, especially when the boundaries between phenomenon and context are

not clearly evident" (p. 18). Because the needs of rural teachers are inextricably linked to the context in which they work, and in which teacher PD takes place, careful in-depth examination within individual rural contexts is an effective way to examine the broader issue. Moreover, multicase study allows the researcher to examine phenomena that have many parts (Stake, 2006), making it an appropriate choice for examining the experiences of multiple sets of rural educators engaging in a PD workshop in different contexts.

While the participants in the broader, ongoing, multiple case study include rural educators from several communities/contexts in Manitoba, the participants in the single case reported on in this paper were made up of administrators (superintendent and principals), teachers involved in a divisional PD committee, and personnel from student services (including the student services coordinator) in a very small school division in southern Manitoba. Of the twelve people that engaged in the half-day PD workshop, five elected to become participants in the research study by filling in the online questionnaire. Two of the five questionnaire respondents also elected to participate in the follow-up, semi-structured interview with the researchers approximately seven weeks after their workshop experience.

Data collection and analysis. The study was approved by the Brandon University Research Ethics Committee (BUREC). Data sources for the study included online questionnaires; recorded semi-structured interviews; PD artifacts such as game scoring sheets, facilitator materials, and group discussion notes; and debrief conversations between the researchers that were recorded following each PD workshop, in which they discussed logistical matters about the workshop facilitation, and their impressions about participant engagement and learning. The recorded interviews and debrief conversations were transcribed for analysis.

In terms of the first case that is the subject of this paper, the following data sources were collected: five participant questionnaires, two interview transcripts, the transcript of the debrief conversation held between the researchers, and a variety of PD artifacts from the workshop (scoring sheets, facilitator materials, notes, etc.). Data were coded thematically and analyzed using what Miles, Huberman, and Saldaña (2014) refer to as a three step process: (1) data condensation, (2) data display, and (3) conclusion drawing/verification (p. 12). Through this process, the researchers engaged in thematic analysis of the data (Hesse-Biber & Leavy, 2006), which allowed the individual case study's findings to emerge.

Findings from Case 1

What participants learned about the change process through the PD simulation.

When asked what they had learned about the change process as a result of participating in the PD simulation, participants made a number of different observations. They identified learning that the change process is complex and that it takes time to produce results. It is important, therefore, to choose change initiatives that will have a high impact, and to start small by building everyone's background knowledge. They remarked on how important it is to inform, include and reach all staff members, and the need to "catch new teachers up" when they join the staff after an initiative is already underway. The rate of change, they found, is impacted by the leader's skill set and the available resources such as time and money. Individual teachers will each take a different path through the change process, so the focus of the change agents will vary per person and shift at different points in the change process:

> Everyone moves through the process differently. Sometimes our efforts can be focused solely on those who are most open to change or alternatively, on those who are most resistant to change. It is important to make sure we are trying to reach everyone. Building

> capacity in other teachers / team members to help lead the change greatly speeds up the process.

To accelerate the rate of change, participants found it was wise to invest time and energy in developing the leadership capacity of team and staff members. Overall, participants reported that, through the simulation, they had increased their understanding of change and the need to consider the needs and responses of individual teachers throughout the change process.

How participants learned through different elements of the PD simulation. Specific elements of the simulation contributed to participants' learning. Working together in collaborative teams with a shared goal was perceived as a supportive feature; while the competitive nature of the game provoked a "desire to succeed" (Questionnaire response). The fast pace of the game enhanced engagement and helped sustain participants' attention. Some participants gained insight from observing one another's responses when proposed activities were unsuccessful. One participant commented on the value of hearing about the different adopter types in the debriefing discussion:

> I especially found the information at the end of the simulation interesting, when the different teacher profiles were categorized into personality types. This is great information to have when it comes to knowing how to motivate and encourage the different staff on your team.

Overall, the game-like structure of the simulation, which required teams to work together to achieve positive outcomes and outpace their rivals, enhanced motivation and engagement; while the research embedded content led to new insights about the change process.

How participants reacted to each phase of the simulation. Participants demonstrated a range of reactions during each phase of the simulation. While the game was being introduced, participants indicated engagement and focus through attentive listening and lowering their heads or closing their eyes in concentration. As the game was explained there was an observable sense of curiosity and excitement in the room about the challenge they were facing. Friendly banter between the teams provided a competitive element between the teams.

As teams moved into the first phase of the game, which represents year one of two for the school PD committee, different types of strategies and reactions were observable. Teams approached their task in different ways. One team read all their material very thoroughly and seemed to want to plan out the whole change process before beginning to take action. A second team jumped quickly into action, selecting activities without any advance planning. This team quickly ran out of tokens, which represent money in the simulation. Participants demonstrated curiosity by asking out loud how they could get more of the four colours of cubes that are used in the simulation to represent capacity building (colour-coded cubes represent leadership, teacher learning, quality teaching and student learning). When teams did start to receive the quality teaching and student learning cubes later on in the game, they started making connections between their choice of selected activities and capacity building on staff and ultimately among students. As playing continued, participants made comments about "silly principals" who think they know everything. This comment was made in reaction to discovering that to be successful in some activities, teams needed to consult the principals at least three times to move ahead in the game. Another connection in phase one was when teams were heard saying "We need to celebrate," a strategy which earned them bonus tokens and a success card each time they chose that option in the game. Similarly, when teams obtain their PD Plan they are rewarded with

additional new tokens, a refund of their expended budget and gain access to a larger list of professional learning activities. As soon as one team was successful in earning their PD plan the curiosity of the other teams was peaked and engagement appeared to increase.

Phase Two of the simulation represents year two for the PD committee and participants' engagement and excitement increased as teachers started to move forward at a faster rate through the selected activities. At this stage of the game, participants expressed curiosity about the progress of individual teachers and could be heard discussing specific teachers by name, specifically Houston and Jada.

The final phase of the simulation included the debriefing session and reflective comments made by the participants at the conclusion of the simulation. Teams continued to remark on the varying rates of progress of the various teachers in the game and adopted different strategies to accelerate their movement. They noticed that gaps were emerging between the teachers in the fictional school, which made them become very focused on accelerating specific teachers' movement around the board. As they compared notes on the approaches they had used to move teachers forward, one team in particular reported focusing specifically and deliberately on the needs of new teachers because they felt that was often not done very well: "I believe it's the right choice to mentor everyone and I think that means new people, and if you skip some teachers in the process then your change doesn't work." Another team reported focusing on those teachers in the middle of the pack and their top teachers, which gradually created momentum that moved their resisters forward. In acknowledging the impact of their strategy one team member remarked that they got the teacher Kanye right to the end of the game, indicative of sustained effective practice. Although there were many observations voiced about the different teachers and the challenges in moving then forward, there was little consensus on which teachers they believed

would fall into the specific categories identified in the debrief as: innovators, leaders, early adopters, middle adopters or resisters (WestEd.com). Many teams reported that the challenges they faced in moving their teachers forward included lack of funds and lack of time.

In their concluding comments after the simulation was over, participants remarked on the different lessons they had learned through their experience. Many participants mentioned learning about the importance of teacher leadership in moving the teachers forward. One person felt the whole simulation was about building teacher capacity, while another reported that sometimes the teachers they perceive as having leadership potential do not want to embrace that role. It was not clear if this comment referred to the teachers in the game or in the participant's school. Another perspective was that it is important to find capacity within your team: "Look for internal leaders, teacher leaders who can drive the bus and look for different people for different initiatives." For one participant their biggest takeaway was the notion of partnering strong and weaker teachers together in a mentoring relationship.

In reflecting on the change process one person's observation was that there are no right or wrong answers, while another person said actually they had learned that it was really important to do your research before launching in blindly without a plan, because some strategies and sequences are more effective than others. Consistency and embracing a shared vision were clearly understood to be key elements in fostering change, as one participant posed the question: "How do we get everyone moving together with a shared vision?" Change was described as a process or a journey instead of an event, "PD is not a one-off and persistence is really important." The importance of strong communication with staff members surfaced as a key takeaway for one participant, while another announced that they had learned it was okay to celebrate success, and that schools do not need to be constantly innovating.

Participants demonstrated varied reactions throughout the different phases of the game. While initially each team adopted a different approach to the task, the process of playing alongside different teams allowed group members to learn from each other to select activities that had proven effective for other teams/committees. Energy and engagement peaked when any team was successful in obtaining new coloured cubes, earning bonus tokens or moving teachers along the path to sustained effective practice. This energy increased in the second phase of the game when the change was gathering momentum indicated by the teachers advancing more rapidly on the game board path towards Sustained Effective Practice. During the debrief, participants commented on the diverse approaches teams used to move teachers forward and made personal connections to their own work context. Finally, in reflecting on their experience with the simulation, participants identified specific learning in the areas of teacher leadership, the nature of change, the importance of a shared vision, communication and the need for frequent celebrations of progress.

Applying the experience to individual contexts. When asked about the connections the participants saw between their experiences engaging in the game board simulation and the educational contexts in which they worked, several similarities were noticed. Participants noted similar budget and time constraints in their own contexts, particularly in relation to running out of money (in the game and for PD in their division), and in terms of questions about having the necessary time to discuss progress. One of the feedback cards in the game suggested that they needed two hours a week to discuss progress; this was something that sparked conversation during the workshop as the participants discussed the challenges with finding such time to focus on an initiative. In participant conversations, issues with regards to multiple initiatives in the division were discussed, as well as the overburdening of administrators. These conversations

were mirrored in one of the participant questionnaire responses that described the connections they saw between the simulation and their own contexts in the following way: “Similar employee traits, budget constraints and time. Leaders stretched too far to accomplish goals.”

Another connection made between participants and their own educational contexts related to high teacher turnover and the need to support new teachers and new administrators in their division. In the game, after one year, teams were given two new teachers to incorporate into their PD plans. When the player pieces and descriptions of the new teachers were handed out to the teams with the explanation that they had two new teachers in their schools for year two, one of the teams said, “Of course we do.” They were clearly able to relate to teacher turnover in their own contexts, which was evident in this comment. During the debrief session after the game, the challenge of high teacher turnover emerged again in conversation with one principal noting that over half of her staff was new this year. Workshop participants questioned whether they were doing a good job of making sure new teachers were supported in their schools, and whether or not new administrators in the division were supported effectively. As a result of such questions, suggestions were made about paying closer attention to the needs of new teachers and administrators in the division in order to provide them with better support.

A third connection made by workshop participants between their experience with the simulation and their own educational contexts was the similarity between the traits of teachers in the game and the traits of teachers in their own school or division. One participant noted on the questionnaire, “This game nailed it for me. [It] was exactly what I see on a regular basis with my staff at present,” while another said, “The teacher profiles were bang on some of [the] actual personalities that I see in fellow educators!” In a follow-up interview a few weeks after the PD session, one of the principals in the division noted the following:

> We're always trying to elicit change in one way or another, here, for sure, and we definitely have teachers on different positions along the spectrum. We have, obviously, some, the people that are just – they buy in right off the bat. They're the real eager ones that just – they're all about it. They're on the bus; they're drinking the Kool-Aid, more or less. You know, they just want to go, they want to progress. Then we have some others that wish to kind of see it in action, and then they'll still be willing to come along. Then . . . if we do have resistors on our staff when we're trying to implement change, I find as a leader, I tend to focus on those ones, and I really have to make sure that I don't lose focus on the rest of the crew who is being successful, and make sure I really say, and praise, what they're doing; and continue to focus and help them on their journey, instead of putting all my resources behind the ones that are being the resistors. And that's why this game really hit home for me, is because we live it here every day. (Participant Interview)

The participants in the study not only recognized similarities between the adopter types in the game and their own colleagues and staffs, they also made connections between the events in the game and their own approaches to dealing with a variety of adopter types in their own contexts.

A fourth connection that was made between the game and the educational contexts in which participants worked, had to do with PD approaches and units of focus. Although the game board simulation focused heavily on teachers in a particular fictitious school, several of the workshop participants noted that divisional PD within their own context(s) was often more focused on topics or activities than it was on people or needs. One participant said the following: "Our experience with Division-wide PD has been activity-focused rather than on long-term,

cultural change that impact[s] student learning. This simulation helped our leaders understand their role[s] in facilitating that change." Conversations following the simulation explored the differences between the teacher-focused approach to PD in the simulation, and divisional approaches to PD, pointing out the division's tendency to rush into doing PD before having important conversations with teachers, and the need to focus on teacher needs, rather than topics, activities, or speakers. One participant in particular noted that they planned to, "ensure everyone has a voice at the PD committee as to the direction things are headed." This comment, along with the ones about ensuring conversations take place with teachers, suggest that there was some interest (after the simulation experience) in being more thoughtful about approaches to PD, and the unit of focus in such approaches.

A final theme that emerged in relation to connections between the simulation and the context of participants had to do with scale. While the simulation focused on the context of a single school attempting to help several teachers improve science instruction, the context of the school division in which the workshop was held was much smaller. Schools in the division were nowhere near large enough to have a science initiative involving so many teachers (they likely only had one or two science teachers in a high school), and as such, the context was not similar to that of the participants in the study. The result of the difference in scale between the simulation context and the divisional context was a weaving in and out of school and divisional contexts during discussions. Participants alternated between making connections to the division, and making connections to their schools, changing fluidly back and forth in discussions. Similarly, responses on the questionnaire also moved back and forth between contextual levels. One participant said the following in relation to connections they saw between their own educational contexts and the simulation: "It was valuable at the school level when we are

working on school-specific PD as well as at the divisional level." While this participant was able to separate the two levels of applicability, it is noteworthy that they had to make connections at both levels of their own context, in order to connect to elements of the game board simulation (set up at the school level).

How the PD simulation helped participants develop leadership capacity. When participants were asked how the simulation helped them to develop their leadership capacity, participants reported feeling an increased clarity around the leadership roles required of a school leader. One school leader commented: "Learned that I must be a persistent advocate, involved participant, cheerleader, party planner, and effective communicator!" For another participant the experience led to increased feelings of confidence. This individual remarked that the simulation experience, "May make me more likely to speak up at meetings". This acknowledgement suggested that the simulation had increased this individual's willingness to voice a personal opinion in different forums. Two participants remarked on how the simulation experience would alter their approach to facilitating change. One had decided to pay less attention to those teachers who were resisting the change; while the second decided to focus more on developing teacher leadership rather than trying to spearhead all change personally.

Not everyone, however reported increased feelings of competence as a leader. For one individual it was a somewhat stressful experience: "I will be honest in that it made me doubt my leadership abilities. Although a simulation, I felt frustrated at all times and realize this frustration is a daily thing." For this individual, the simulation evoked feelings of frustration, self-doubt and a lack of confidence in their ability to facilitate change in their current context. The third type of comment from participants indicated that they had gained clarity, either about the purpose of professional development being to plan for student learning, or about the nature of change: "This

experience provided me with a better understanding of the change process and all of the elements that must come together to produce positive results".

Although most respondents reported increased clarity and insight into their roles, responsibilities and options as leaders, not everyone felt an increased sense of efficacy as a result of their participation in the simulation. Leadership capacity was enhanced because participants increased their understanding of the purpose of professional learning and elements of the change process.

Significance of the Study

A consistent theme throughout the literature is the scarcity of opportunities for teacher leaders to develop leadership capacity through professional learning (Katzenmeyer & Moller, 2009; Killion et al., 2016; Lambert, Zimmerman & Gordon, 2016; Levin & Schrum, 2017; York-Barr & Duke, 2004). Leading adults in a process of change requires, at a bare minimum, knowledge of adult learners, facilitation skills, and the ability to engage in and model reflective practice and metacognition (Levin & Schrum, 2017). Through this simulation board game, participating educators were able to vicariously experience a change process, reflect on what they learned in the process and on how the process would need to be different in their own rural context. Feedback from participants indicated that they gained an appreciation for the micro-politics of making change in school settings, the variables caused by individual responses, the need for multiple approaches, the need to celebrate small shifts in practice and the importance of a sustained focus. The experience of navigating a change process with other team members allowed participants to experience various ways of responding to resistance and conflict. As a result they commented that they would be less worried about moving the resisters forward in the future, and focus instead on developing teacher leaders who could champion the desired change.

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